EDMS 04−100

SPECIFICATION

FOR

OVERHEAD ALUMINUM CONDUCTORS

STEEL REINFORCED (ACSR)

Issue: Jul−2020 / Rev− 1

This revision contains option items that may be requested
by distribution company before bidding

CONTENTS

1- SCOPE

This specification describes the minimum technical requirements for the design, manufacture, factory test, supply and delivery of bare concentric-lay stranded Aluminum Conductor Steel Reinforced (ACSR). The conductor shall be suitable for use in the outdoor construction of overhead power distribution network of Egyptian Electricity Distribution Companies (EDC). The conductor is intended to carry the electrical power over the long distance medium voltage MV (12 kV and 24 kV, 50 Hz) distribution networks.

2- APPLICABLE CODES and STANDARDS

The equipment/material covered in this specification shall comply with the latest edition/amendment of the following codes and standards. Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply.

- BS EN 50182 Conductors for overhead lines. Round wire stranded conductors.
- BS EN 50189 Conductors for overhead lines. Zinc coated steel wires.
- BS EN 2004 Test Methods for Aluminum and Aluminum Alloy products.
- IEC 61089 Round wire concentric lay overhead electrical stranded conductors.
- IEC 60888 Zinc-coated steel wires for stranded conductors.
- IEC 60889 Hard-drawn aluminum wire for overhead line conductors.
- IEC 61597 Overhead conductors – Calculation methods for stranded conductors.
- IEC 60468 Method of measurement of resistivity of metallic materials.
- IEC 61394 Overhead lines – Characteristics of greases for bare conductors.
- IEEE 738 Calculating the current-temperature relationship of bare conductors.

- BS 215-1 Aluminum conductors and aluminum conductors steel reinforced for overhead power transmission. Part I: Aluminum stranded conductors.
- BS 215-2 Aluminum conductors and aluminum conductors steel reinforced for overhead power transmission. Part II: Aluminum conductors, steel-reinforced.
- ASTM B- 232 Concentric-Lay-Stranded Aluminum Conductors, Coated-Steel Reinforced (ACSR).

3- SERVICE CONDITIONS

The conductor is required to have a good electrical and mechanical performance under the following local service conditions:

Mean daily temperature	40 °c
Maximum ambient shade temperature	50 °c
Minimum ambient shade temperature	-10 °c
Maximum relative humidity	95 %
Environmental conditions	Humid tropical climate with heavily polluted atmosphere
Maximum Rainfall	250 mm
Altitude	Up to 1000 m above sea-level

4- DESIGN and CONSTRUCTION REQUIREMENTS

4.1 General:

- The ACSR conductor shall meet the requirements of this specification in all respects. Manufacturer’s drawings shall show the cross-section of the bare overhead line conductor, together with all pertinent dimensions

4.2 Construction:

- The conductor shall consist of a solid or stranded round zinc coated galvanized steel core surrounded by strands of hard-drawn round aluminum wires.
- The bare overhead line conductor shall be constructed in conventional concentric-lay conductor type. The direction of lay of the outer layer shall be right hand and shall be reversed in successive layers.
- The lay ratio of the stranded steel layer shall be (16 to 26). Whereas for aluminum layers, the lay ratio shall be (10 to 14) for the outer layer and (10 to 16) for all inner layers.
- The surface of the wire shall be smooth and free from imperfections not consistent with good manufacturing practice.

4.3 Design Criteria:

- Unless otherwise specified, the conductor shall be designed to fulfill the electrical and mechanical requirements of the relevant standards in clause 2.
- The conductor shall be designed for service conditions specified in clause 3.
- The size and continuous current carrying capacity of the conductor shall be based on a maximum permissible continuous temperature of 90°C taking into account the solar radiation and wind effects.
- The ACSR conductor rating and dimensions shall be as indicated in Table (1).
- The rated strength given in Table (1) for a specific conductor is taken as the sum of the strengths of the aluminum and steel components.
- The breaking tensile strength of the composite conductor after stranding shall not be less than 95% of the appropriate rated strength.
- The coefficient of linear expansion and the final modulus of elasticity of the

composite conductor shall be as follows:

No. of conductor wires	Coefficient of linear expansion	Final modulus of Elasticity
06 AL. / 1 Steel	$\leq 1.92\times10\text{-}5\ K^{-1}$	81 kN/mm^2
26 AL. / 7 Steel	$\leq 1.90\times10\text{-}5\ K^{-1}$	77 kN/mm^2

- No joints of any kind shall be made in the finished steel wires. Joints may be made in the rods or semi-finished wires prior to drawing to final size, provided that the supplier can guarantee that the joint will endure at least 90% of the tensile strength of the unjointed rod.
- Welded joints in the aluminum wires shall not be closer than 15 m to another or to either end of the wire. No more than two such joints shall be present in any reel length of the conductor.

4.4 Area and Diameter:

- The area and diameter of bare overhead line conductor covered by this specification are given in Table (1)

4.5 Mass and Electrical Resistance:

- The appropriate mass and electrical DC resistance for ACSR at 20°C given in Table (1) are subjected to the standard increments due to stranding shown in Table (2).

4.6 Grease (Optional):

- The contractor is required to smear a number of the conductor inner layers (determined by _ _ EDC) with an approved anti-corrosive grease having a high melting point and complying with IEC 61394.
- The mass of grease shall not vary by more than 20% from the calculated value obtained using the method described in annex B of BS EN 50182.

Table (1): Characteristics of Aluminum Conductors Steel Reinforced (ACSR)

Code'	Old code	Areas			No. of wires		Wire diameter		Diameter		Mass per unit length	Rated Strength	DC Resistance	Current Carrying Capacity
		AL.	Steel	Total			AL.	Steel	Core	Cond.				
		mm^2	mm^2	mm^2	AL.	Steel	mm	mm	mm	mm	kg/km	kN	Ω/km	A
15-AL1/3-ST1A	16/2,5	15,3	2,54	17,8	6	1	1,80	1,80	1,80	5,40	61,6	5,80	1,876 9	105
24-AL1/4-ST1A	25/4	23,9	3,98	27,8	6	1	2,25	2,25	2,25	6,75	96,3	8,95	1,201 2	140
34-AL1/6-ST1A	35/6	34,4	5,73	40,1	6	1	2,70	2,70	2,70	8,10	138,7	12,37	0,834 2	170
48-AL1/8-ST1A	50/8	48,3	8,04	56,3	6	1	3,20	3,20	3,20	9,60	194, 8	16,81	0.593 9	210
70-AL1/11-ST1A	70/12	69,9	11,4	81,3	26	7	1,85	1,44	4,32	11,7	282,2	26,27	0,413 2	290
94-AL1/15-ST1A	95/15	94,4	15,3	109,7	26	7	2,15	1,67	5,01	13,6	380,6	34,93	0,306 0	350
122-AL1/20-ST1A	120/20	121,6	19,8	141,4	26	7	2,44	1,90	5,70	15,5	491,0	44,50	0,237 6	410
149-AL1/24-ST1A	150/25	148,9	24,2	173,1	26	7	2,70	2,10	6,30	17,1	600,8	53,67	0,194 0	470
184-AL1/30-ST1A	185/30	183,8	29,8	213,6	26	7	3,00	2,33	6,99	19,0	741,0	65,27	0,157 1	535
209-AL1/34-ST1A	210/35	209,1	34,1	243,2	26	7	3,20	2,49	7,47	20,3	844,1	73,36	0,138 1	590
243-AL1/39-ST1A	240/40	243,1	39,5	282,5	26	7	3,45	2,68	8,04	21,8	980,1	85,12	0,118 8	645
304-AL1/49-ST1A	300/50	304,3	49,5	353,7	26	7	3,86	3,00	9,00	24,4	1227,3	105,09	0,094 9	740
NOTE 1	Direction of lay of external layer is right-hand (Z)													
NOTE 2	Guidcline values of currnnt carrying capacity arc valid up to a frequency of 60 Hz, assuming a wind vclocity of 0,6 *m/s*, an initial ambicnt temperature of 35 •c and a conductor temperature of 80 "C. For special applications, when there is no air turbulence, the values should be reduced by 30 %.													

Table (2): Increments due to stranding

Stranding of conductor				Increment (Increase) (%)			
Aluminum		Steel		Mass		Electrical resistance	
No. of wires	No. of layers *	No. of wires	No. of layers *	AL.	Zn coated steel	AL.	steel
6	1	1	-	1,39	-	1,39	-
26	2	7	1	2, 18	0,52	2,18	0,52
* Number of layers of each type of wire, not including the center wire.							

- The grease shall completely fill the interstices between the appropriate strands. There shall be no excess grease remaining on the outer surface, which might cause adherence of grit and other material during conductor pulling out and erection.
- The minimum drop point of the grease shall be 120°C, and it shall not migrate towards the bottom of the conductor when the conductor is maintained, continuously, at a temperature of 95°C.
- The grease shall retain its properties at all operating temperatures between -10°C and +100°C. The specified characteristics of the grease shall be unimpaired after heating to 15°C above its drop point for 150 hours. The grease shall not flow within, nor exude from the conductor, at temperatures up to and including 120°C.
- Mixing different greases in any one length of conductor shall not be permitted.
- The grease shall protect the conductor from corrosion in service, which may include operation in atmospheres containing salt spray or industrial pollution.
- The grease shall not corrode the component wires of the conductor.

- The grease shall be compatible with any lubricant commonly used during wire drawing that may be present in the conductor strands.
- The grease must not present a health hazard and shall meet all relevant current health safety requirements.
- The Contractor shall give details of the control, supervision and testing measures taken to ensure that there are no discontinuities in greasing.

5- TOLERANCES on NOMINAL DIAMETERS of WIRES

5-1 Aluminum wires:

- The aluminum wires shall not depart from the diameter by more than the following amounts:-

Nominal diameter	Tolerance
2.5 mm and greater	± 1%
Less than 2.50 mm	± 0.025 mm

5-2 Zinc-Coated steel wires:

- The Zinc-coated steel wires shall not depart from the nominal diameter by more than the following amounts:-

Nominal diameter	Tolerance
2.0 mm and greater	± 2%
Less than 2.00 mm	± 0.04 mm

- The diameter shall be measured over the Zinc-coating.
- Each conductor is to be gauged at three places, one near each end and one approximately in the middle. _ _ EDC has the right to reject the conductor if one of the three places is off-gauge by more than the allowed tolerance.

6- TESTING

6.1 General

- Tests shall be carried out on wire samples and on finished conductor in accordance with the latest relevant standard and as specified herein.
- Tests shall be carried out in the presence and under the control of the inspecting engineer of _ _ EDC at work, except in specified or approved cases where test certificates are accepted. All test results shall be reviewed and approved in writing by _ _ EDC.
- All tests shall be carried out by and at the expense of the tenderer who shall supply all the pieces and specimens as well as all apparatus, instruments and equipment.
- Not less than two weeks notice of all tests shall be given. The tenderer shall not be entitled to any extension of the time of completion due to the failure of any test or the rejection of any part of the materials as a result of any test.
- The full range of type and sample tests specified in the relevant standard shall be carried out as applicable.
- Sample tests shall be carried out in the supplier's factory or at any other mutually approved place. Type test report/certificate from an independent testing laboratory shall be submitted to _ _ EDC.

6.2 Samples size

- Tests on wire samples may be carried out before or after stranding. In the case of finished conductors, tests may be replaced by calculations based on the test results on the individual wires. The Contractor must furnish these calculations.
- Tests shall be carried out on a minimum of 10% of the drums offered for inspection and, in such cases, each wire shall be tested. Drums to be sampled shall be selected at

random and samples taken from the outer end of the drums.

- The length of the sample of the conductor taken shall be sufficient to allow all tests to be performed on the same specimen of wire.
- In order to check the grease, a sample of the conductor shall be taken from one drum of each inspection lot.

6.3 Acceptance / Rejection

- Failure of a test specimen to comply with any one of the requirements of this specification shall constitute grounds for rejection of the lot represented by the specimen.
- If any lot so rejected, the supplier shall have the right to retest once all individual drums of conductor in the defected lot and submit those, which meet the requirements, for acceptance.

6.4 Sample tests

- The sample tests listed in table (3) are intended to guarantee the quality of conductors and compliance with the requirements of this specification.

Table (3): Sample tests for ACSR

Sample tests for Aluminum wire	Sample tests for Steel wire	Sample tests for complete conductor
• Diameter measurement • Tensile strength test • Elongation test • Resistivity test • Wrapping test	• Diameter • Tensile strength • Stress at 1% extension • Elongation or torsion test • Wrapping test • Mass of zinc • Zinc dip test • Adhesion of zinc coating	• Surface condition • Diameter • Inertness • Lay ratio and direction of lay • Number and type of wires • Mass per unit length

- Grease (if required) shall be tested for mass per unit length and drop point.
- Records for all joints made in the conductor wires shall be submitted to _ _ EDC.

7- PACKING and MARKING

- The conductor shall be delivered on standard sized nonreturnable wooden drums of sturdy construction properly packed and lagged externally to prevent possible damage to the conductor during transportation or outdoor storage.
- Drums shall be made of well-seasoned soft wood free from defects that may materially weaken the component parts of the drums. Wood shall be treated to prevent deterioration by termites or fungus attack. The chemical used for treating the wood shall not be harmful to the conductor.
- External flange diameter shall be such that the distance between the outer edge of the flange and the packed conductor shall be not less than 75 mm. Drums shall be suitable for rolling on the flanges without causing damage to the conductor and the direction of rolling shall be clearly shown.
- All drums shall have spindle hole of adequate diameter and shall be reinforced with steel plates and by bolts and shall be designed for mounting on a horizontal axle for laying out the conductor. The hub and the flanges of the drum shall be securely fastened together with reinforcing by bolts & nuts.
- Wood lagging shall also be secured with steel straps to provide physical protection for the conductor during transit and during customary storage and handling operations.
- The packing shall provide suitable protection against all climatic conditions prevailing during transport and on site. Each end of conductor shall be durably sealed before shipment to prevent ingress of moisture.

- Conductors shall be delivered in drums with conductor lengths of 1 km. Larger drum lengths are acceptable provided that the gross weight of any drum with conductor shall not exceed 2000 kg. These lengths shall be subject to a permitted tolerance of ± 5%.
- Drums shall be marked in legible and indelible letters on a metal plate, affixed to it in a non detachable manner, giving the following particulars:
    1. Conductor Code and type
    2. Cross-section of conductor
    3. Serial number of reel
    4. EDC name
    5. EDC address and purchase order number
    6. EDC stock number in 10 cm high bold numerals
    7. Manufacturer's name and country of origin
    8. Length and weight of conductor on reel
    9. Stranding
    10. Gross weight
    11. Size of reel
    12. Year of manufacture
    13. Direction of rolling of reel
- All markings shall appear on both sides of the reel.

8- GUARANTEE

- The supplier guarantee the conductor against all defects arising out of faulty design or workmanship, or of defective material for a period of two years from date of delivery.

9- SUBMITTALS

9.1 Submittals required with tender:

- The supplier shall complete and return one copy of the attached Data Schedule for each size of conductor offered.
- Technical data and original catalogues shall be submitted to facilitate evaluation of the offer.
- Dimensional cross-sectional drawings of conductors and drums.
- Approval certificate issued by Egyptian Electricity Holding Company.
- Type test certificates.
- List of deviations & clauses to which exception is taken (if any).
- Guaranteed Ex-Works delivery date.

9.2 Submittals required following award of contract:

- Details of manufacturing and test programs.
- Factory test reports.

10- TECHNICAL DATA SCHEDULE

- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.

SCHEDULE OF PARTICULARS

TECHNICAL DATA of ACSR CONDUCTOR

_ _ EDC Inquiry No:______________________________ Item No:________________

(Sheet 1/2)

No.	DESCRIPTION	UNIT	_ _ EDC Specified Values*	Vendor Proposed Values**
2.0	The standard to which cable or wire manufactured	As per clause 2		
4.0	DESIGN and CONSTRUCTION REQUIREMENTS			
1	Type of conductor	ACSR		
2	Number and diameter of aluminum strands	No./mm	As per table1	
3	Aluminum area	mm^2	As per table1	
4	Number and diameter of steel strand	No./mm	As per table1	
5	Steel area	mm^2	As per table1	
6	Total cross-sectional area of conductor	mm^2	As per table1	
7	Overall diameter of bare conductor (apprx.)	mm	As per table1	
8	Ultimate breaking load of conductor	kN	As per table1	
9	Rated tensile strength	kN	As per table1	
10	Equivalent modulus of elasticity	kg/mm^2	Clause 4.3	
11	Maximum stress	kg/mm^2	Clause 4.3	
12	Equivalent coefficient of linear expansion per °C	1/K	Clause 4.3	
13	Max. continuous current carrying capacity of conductor at ambient temperature 35°C	Amps	As per table1	
14	Conductor temperature for condition at item 13	° C		
15	The short time overload current capacity of conductor at ambient 50°C - 1 hour duration - 3 hour duration	Amps Amps		
16	Conductor temperature for condition at item 15	° C		

17	Max DC resistance per km at 20°C	Ω/km	As per table1	
18	Max AC resistance per km at 80°C	Ω/km		
19	Weight of conductor per km	kg/km	As per table1	
20	Greased layers and Grease Name (if any)	-	Clause 4.6	
21	Minimum weight of grease (if any)	kg/km	Clause 4.6	

(Sheet 2/2)

No.	DESCRIPTION	_ _ EDC Specified Values*	Vendor Proposed Values**
7.0	PACKING and MARKING		
1	Drum type	Wooden	
2	Length of conductor (m)	As per clause 7	
3	Dimensions (m)	As per clause 7	
4	Gross weight (kg)	As per clause 7	
5	Net weight (kg)	As per clause 7	
6	Marking as per the specification	Yes	
9.0	SUBMITTALS		
1	All submitted as per the specification	Yes	

(*) - Values to be provided / proposed by the vendor.
(**) – Please, provide explanations for deviations if any.

I / We guarantee the technical date given above for the offered material/equipment.

Description	Manufacturer of Material/Equipment	Vendor/Supplier
Vendor/Supplier		
Location and Office Address		
Name and Signature of Authorized Representative with Date		
Official Seal / Stamp		